Daniel Lee

408-826-2633 | danlee4293@gmail.com | linkedin.com/in/lee4293 | github.com/lee4293

Education

University of California, Davis — Davis, CA
Master of Science in Computer Science — *September 2026 – June 2028*

Purdue University — West Lafayette, IN
Bachelor of Science in Data Science — *August 2022 – May 2026*

Work Experience

Software Engineer Intern — July 2024 – October 2024
Daedalus Labs — *San Francisco, CA*

- Built a **Node.js** ingestion pipeline on **Microsoft Azure** to standardize fragmented blockchain webhook payloads, persisting **10,000+** Bitcoin Rune transactions in **Azure SQL** for reliable transaction monitoring
- Developed a near-real-time monitoring system for Bitcoin transactions and Nostr protocol events, delivering alerts through Discord webhooks and a Chrome extension
- Evaluated Fractal Bitcoin's transaction model using protocol documentation and transaction-level data, translating findings into monitoring and ingestion requirements during **Agile** sprint reviews

Data Analyst Intern — August 2023 – May 2024
NSF I-GUIDE — *West Lafayette, IN*

- Integrated flood-inundation rasters from **345+ dams** with census-block socioeconomic data using **Python** to assess population exposure and vulnerability across flood-prone regions
- Containerized and distributed geospatial pipelines using **Docker**, **Apache Spark**, and **Apache Sedona**, standardizing local and **HPC** environments while scaling raster-vector operations and large-area spatial joins
- Profiled raster-heavy Python workloads using timing and memory diagnostics, isolating computational and data-access bottlenecks to target high-cost pipeline stages

Research and Publications

Undergraduate Student Research — August 2022 – May 2023
The Data Mine (Purdue University) — *West Lafayette, IN*

- Investigated rising pediatric melatonin ingestion incidents with the **CDC PROTECT initiative**, analyzing product labels and online marketplace descriptions to identify potential sources of consumer misunderstanding and accidental misuse
- Filtered and standardized labeling data for **347+** melatonin products from the NIH Dietary Supplement Label Database, creating a reference schema for comparing ingredient, dosage, warning, and marketing language across products
- Built **NLP** pipelines with **Hugging Face Transformers** to perform topic modeling and sentiment analysis on **3,500+ Amazon reviews** and **410+ Reddit posts**, surfacing recurring consumer concerns, misconceptions, and usage patterns to inform the broader labeling analysis

Projects

Nuveri.app | *TypeScript, Next.js, React, Anthropic API, Supabase* — August 2026 – Present

- Built a full-stack research platform with **Next.js, TypeScript, and Supabase/PostgreSQL**, aggregating scholarly data to let users search, evaluate, and trace scientific claims back to primary research
- Engineered a **TypeScript** and **Anthropic** analysis pipeline combining LLM inference with deterministic validation to produce source-grounded research assessments while reducing unsupported conclusions

LEO Satellite Telemetry Anomaly Detection | *Python, scikit-learn, Pandas* — January 2026 – May 2026

- Built a reproducible **Python** and **scikit-learn** anomaly-detection pipeline, engineering 28 statistical and channel features from **2,123 telemetry segments** and benchmarking supervised models to identify satellite outages, interference, and link failures
- Extended detection to unlabeled orbital data by engineering 18 features from **9,919 Starlink records** and ensembling Isolation Forest with One-Class SVM to flag potential position anomalies and unexpected orbital behavior without ground-truth labels

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL
Frameworks/Libraries: React, Next.js, Node.js, scikit-learn, Pandas, Apache Spark, Apache Sedona, Hugging Face Transformers, Tailwind CSS
Databases/Cloud/Tools: PostgreSQL, Azure SQL, Microsoft Azure, Docker, Git, Jupyter